

A support-based obstruction to termwise Bombieri–Vinogradov extraction in a Möbius–prime convolution

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Abstract

Let N be a large even integer and $C(N) = \sum_{n < N} \Lambda(n)\mu(N-n)$. In the Huang–Li reduction of binary Goldbach to Elliott–Halberstam [4], two related Möbius-weighted fixed-class sums occur: a flat branch, which is unconditionally small in the regime of [9], and a logarithmic branch tied to Huang–Li’s residual relation. This raises a natural design question: can an intermediate weight make the divisor-switched residual accessible to Bombieri–Vinogradov while retaining a non-negligible coefficient of $C(N)$?

For the standard *termwise* Bombieri–Vinogradov evaluation, the answer is no on the stated low-transform support class. Write $b = \mu * w$. If b is supported on $d \leq N^{\theta' - 1/2 - \delta}$, then the residual is controlled termwise, but the extraction coefficient satisfies

$$\frac{\|b\|_1}{|B_w|} \gg \exp\left(c_1 \sqrt{(\tfrac{1}{2} + \delta) \log N}\right).$$

The loss exceeds every fixed power of $\log N$. The same support-based triangle-inequality argument under a level $EH(N^{\theta_E})$, with fixed $1 - \theta' < \theta_E < 1$ and an arbitrarily small fixed power slack in the support, retains an exponential loss at least as large as the one governed by $N^{1-\theta_E}$.

The scope is essential. The support condition is sufficient for the termwise progression estimate; it is not proved necessary, and cancellation among the b_d or among the three terms of the switch identity is not excluded. Polynomial weights lie outside the support class and are analysed only structurally: monomials give nonnegative complete parts, while general tuning leads back to prime-pair moments not estimated here. No impossibility theorem is claimed for all polynomial weights.

The arbitrary-weight switch identity needed for the no-go is proved directly here, so the main theorem does not depend on importing an unstated generalisation from the companion paper. Six finite rearrangements describe the surrounding design space; the appendix states precisely which algebraic scaffolding is formalised in Lean 4. No new bound for $C(N)$ or for Goldbach is obtained. The result closes one explicit triangle-inequality evaluation, not the underlying correlation problem.

Keywords: divisor switching; Bombieri–Vinogradov theorem; Möbius function; Elliott–Halberstam conjecture; Goldbach problem.

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1 Introduction

1.1 The setting

Let N be a large even integer, $\theta' \in (1/2, 1)$ fixed, and $K = N^{\theta'}$. Put

$$C(N) = \sum_{n < N} \Lambda(n)\mu(N-n), \quad A(N; k) = \sum_{\substack{n < N \\ n \equiv N \pmod{k}}} \Lambda(n)\mu(N-n), \quad (1)$$

and

$$\tilde{r}(N) := \sum_{n < N} \Lambda(n) \Lambda(N - n), \quad r_{\mathbb{P}}(N) := \sum_{\substack{p+q=N \\ p, q \text{ prime}}} \log p \log q.$$

The first is the standard von-Mangoldt convolution, not literally the prime–prime count: the terms in which at least one variable is a proper prime power contribute $O(N^{1/2} \log^2 N)$, so $\tilde{r}(N) = r_{\mathbb{P}}(N) + O(N^{1/2} \log^2 N)$. Thus a positive lower bound for $\tilde{r}(N)$ with margin $\gg N^{1/2} \log^2 N$ implies binary Goldbach, whereas bare positivity of $\tilde{r}(N)$ does not. For a weight $w : \mathbb{N} \rightarrow \mathbb{R}$ put

$$T_w = \sum_{\substack{k < K \\ (k, N)=1}} \mu(k) w_k \left[A(N; k) - \frac{C(N)}{\varphi(k)} \right]. \quad (2)$$

The functional E_3 of [4] is T_w at $w_k = \log k$; E_4 is not T_w at $w_k = 1$ but is obtained from it by partial summation, an inner factor $\log(N - n)$ separating the two [9, Corollary 2]. E_4 is unconditionally $\ll_A N (\log N)^{-A}$, while the E_3 identity is equivalent, at the stated error scale, to Huang–Li’s equation (22) [9]. This paper studies the part of the intervening weight space for which the switch residual can be controlled *termwise* by Bombieri–Vinogradov, and records what remains open. A sufficient support condition for that termwise control is $b = \mu * w$ supported below $N^{\theta' - 1/2 - \delta}$ for some $\delta > 0$. The polynomial family — including $w_k = \log k$ itself — is discussed separately by a weaker structural argument and is not ruled out. Weights that lie neither in the low-support class nor in that polynomial family are not treated here. §1.4 says which is which.

Throughout, φ is Euler’s function, $\text{rad}(u) = \prod_{p|u} p$, $c, c_1, \dots$ are positive constants depending at most on the parameters each statement fixes $(\theta', \delta, \theta_E)$ and not on N , and

$$\mathfrak{A}(N) = \prod_{\substack{p \text{ prime} \\ p \nmid N}} \left(1 - \frac{1}{p(p-1)} \right), \quad \mathfrak{S}(N) = 2 \prod_{p > 2} \left(1 - \frac{1}{(p-1)^2} \right) \prod_{\substack{p|N \\ p > 2}} \left(1 + \frac{1}{p-2} \right). \quad (3)$$

1.2 The design space

Let w be arbitrary and let $b := \mu * w$, so that $w_k = \sum_{d|k} b_d$; every weight has this form, and b is determined by w . Put

$$B_w := \sum_{\substack{k < K \\ (k, N)=1}} \frac{\mu(k) w_k}{\varphi(k)}, \quad \|b\|_1 := \sum_{u < N} |b_u|. \quad (4)$$

The arbitrary-weight switch needed below is short enough to prove here. This removes a dependence that would otherwise be easy to misread: the companion paper [9] proves the flat and logarithmic branches, but does not state the following arbitrary-weight identity as its Theorem C. The external inputs used later are instead the decay $\rho_n(x) \ll e^{-c_1 \sqrt{\log x}}$ from Huang–Li’s Lemma 1, the Bombieri–Vinogradov theorem in the form of [7], the logarithmic residual relation of [9] used in Proposition 13, and Vaughan’s lower bound for $\|S_{\Lambda}\|_1$ [10], used in Proposition 29.

Lemma 1. *For squarefree u ,*

$$\sum_{k|u} \mu(k) w_k = \mu(u) b_u.$$

Proof. Since $w_k = \sum_{d|k} b_d$ and u is squarefree,

$$\sum_{k|u} \mu(k) w_k = \sum_{d|u} b_d \mu(d) \sum_{j|(u/d)} \mu(j) = \mu(u) b_u,$$

using $(d, u/d) = 1$. □

Proposition 2 (the weighted switch). *Put*

$$\mathcal{R}_w := \sum_{u < N} \Lambda(N - u) \mu(u) \sum_{\substack{k|u \\ k \geq K \\ (k, N)=1}} \mu(k) w_k.$$

Then, uniformly for arbitrary real weights,

$$B_w C(N) = \sum_{u < N} \Lambda(N - u) \mu^2(u) b_u - \mathcal{R}_w - T_w + O(\|b\|_1 N^{o(1)}). \quad (5)$$

Moreover

$$\mathcal{R}_w = \sum_{m < N/K} \mu(m) \sum_{\substack{k \geq K, mk < N \\ (k, mN)=1}} \mu^2(k) w_k \Lambda(N - mk),$$

so expanding $w_k = \sum_{d|k} b_d$ gives $\mathcal{R}_w = \sum_d b_d \mathcal{R}^{(d)}$.

Proof. Write

$$D_w := \sum_{\substack{k < K \\ (k, N)=1}} \mu(k) w_k A(N; k), \quad T_w = D_w - B_w C(N).$$

Since $k \mid N - n$ in $A(N; k)$, the finite change of variables $u = N - n$ gives

$$D_w = \sum_{u < N} \Lambda(N - u) \mu(u) \sum_{\substack{k|u \\ k < K \\ (k, N)=1}} \mu(k) w_k.$$

Complete the divisor sum and subtract its tail. The tail is $\mathcal{R}_w$. If $(u, N) = 1$, Lemma 1 turns the complete sum into $\mu(u) b_u$, hence its contribution is $\Lambda(N - u) \mu^2(u) b_u$.

It remains only to account for squarefree u with $(u, N) > 1$. If $\Lambda(N - u) \neq 0$ and a prime p divides (u, N) , then $p \mid N - u$, so $N - u$ must be a power of p . Thus the exceptional u lie among $u = N - p^a$ with $p \mid N$; there are $N^{o(1)}$ of them. For each such u ,

$$\sum_{k|u} |w_k| \leq \sum_{k|u} \sum_{d|k} |b_d| \leq \tau(u) \|b\|_1 = N^{o(1)} \|b\|_1,$$

and $\Lambda(N - u) \ll \log N$. This proves the homogeneous error term in (5). Finally write $u = mk$ in the tail. The factor $\mu(u)$ forces u squarefree, hence $(m, k) = 1$ and $\mu(u) \mu(k) = \mu(m) \mu^2(k)$, giving the displayed formula for $\mathcal{R}_w$. $\square$

The loss theorem below depends most sensitively on the decay of $\rho_n(x)$, not on an unproved generalisation of the switch. The measurement of Lemma 16 checks its factorisation identity for single-divisor weights; it does not verify the analytic decay estimate.

Note 3 (the third term is not a remainder). The term T_w in (5) is (2) itself. It must remain explicit in the structural identity: at $w = \log$ it is E_3 , which [9, Theorem 3] evaluates as $\tilde{r}(N) - \mathfrak{S}(N)(N - C(N)) + O_A(N(\log N)^{-A})$. Deleting it from the identity changes the two sides by T_w itself — the two sides then differ by $0.44N, 0.38N, 0.32N$ at $N = 2 \cdot 10^5, 4 \cdot 10^5, 8 \cdot 10^5$ with $\theta' = 0.56$, which are exactly the measured values of $|T_{\log}|/N$ at those N , read from the run Measurement 8 cites rather than from that measurement's own printed row; over the eight N that run sweeps, the same ratio falls to 0.1245 , so over that range the two sides differ by a decreasing fraction of N and not by a constant multiple of it. That the decrease is a power of N is a fit to those eight points and not a bound; the use made of it here is only the lower one, that $|T_{\log}|$ stays above $0.12N$ throughout. The $w = 1$ case of the same collection,

$T_1 = \sum H - C(N)B(K)$, is checked numerically and holds to a worst relative error $1.875 \cdot 10^{-16}$ (`lab_weight_gap.py`). The homogeneous exceptional-set error in Proposition 2 is deliberately crude for arbitrary weights; for $w = \log$ its norm bound is not the device used to measure $T_{\log}$.

Keeping it explicit is what makes Theorem 20 say something. The identity by itself yields nothing about $C(N)$. The flat weight $w = 1$ is the case for which [9, Theorem 1] supplies the required strong bound on T_w . The theorem below grants the corresponding *homogeneous* bound for a general admissible weight and shows that the stated triangle-inequality extraction still loses all fixed log powers.

The two known cases are the two ends of this space: $w = 1$ gives $b = \delta_1$, $\|b\|_1 = 1$ and (Huang–Li’s Lemma 1) $B_w \ll e^{-c\sqrt{\log K}}$; while $w = \log$ gives $b = \Lambda$, $\|b\|_1 \asymp N$ — the complete part is the prime-supported sum $\sum_{p < N} \Lambda(N - p) \log p$, differing from $\tilde{r}(N)$ only by proper-prime-power terms — and $B_w = -\mathfrak{S}(N) + O(e^{-c\sqrt{\log K}} \log K)$ in the regime used below. For even N , $\mathfrak{S}(N)$ is bounded below by a positive absolute constant (and may grow slowly with the prime factors of N), so this coefficient is not exponentially damped.

1.3 Results

Section 2 records the exact structure of the space: the per-modulus discrepancy is a dilate (Proposition 4), the signs $\mu(k)$ cancel exactly on the $\log k$ branch (Proposition 5), both known weights are the same sum of dilated walls with different weights (Proposition 6), the untruncated logarithmic object is the prime-supported weighted pair sum of Proposition 9, which differs from $\tilde{r}(N)$ only by proper-prime-power terms and is therefore a signed sum of nonnegative layers (Proposition 10) over Bombieri–Vinogradov moduli carrying no Möbius (Proposition 11); and the wall cancels out of the count entirely (Proposition 13).

Section 3 proves the scoped no-go: Lemmas 16 and 19 are the two opposing estimates, Theorem 20 is their consequence for the termwise Bombieri–Vinogradov support class, and Theorem 24 gives the corresponding loss at higher fixed levels of distribution. Section 4 records what the same argument does and does not say about polynomial weights, which lie outside the support hypothesis of Theorem 20. Section 5 records the position of the circle method.

1.4 What is not claimed

- **This is a no-go for one method, and for the part of its weight space that method can evaluate.** Theorem 20 closes the *termwise, triangle-inequality* divisor-switch evaluation on the explicit support class in its statement. It does not prove that this support condition characterises every weight that Bombieri–Vinogradov could conceivably help evaluate. Of the weights outside that hypothesis, the polynomial family $w_k = f(\log k)$ is addressed by Proposition 26, whose (ii) is a statement of what is not known rather than an impossibility; the weights that are neither reached by the hypothesis nor polynomial are closed nowhere here. It is not, and does not claim to be, an obstruction to other methods — in particular it says nothing about the spectral large sieve of [5].
- **The genre is classical.** Bombieri’s asymptotic sieve [2] is a central formulation of the parity obstruction: strong distributional sieve information by itself does not in general distinguish the parity of the number of prime factors. It motivates a statement of this shape, but it is not cited here as a theorem about every possible choice of weight. What is added here is the precise quantitative form for this particular route (Note 23).
- **Proposition 29 is about two pairings, not about the circle method.** What is shown is that the $(2, 2)$ and $(\infty, 1)$ Hölder pairings of the integral lie at or above the trivial bound. A circle-method treatment of a binary problem does not ordinarily proceed by a global pairing at all: it splits into major and minor arcs and treats the minor arcs

bilinearly. The Parseval floor is taken over the whole circle and so does not depend on where that split is drawn, but a bilinear treatment of the minor arcs is a different mechanism and is not addressed here. Nothing below measures the major arcs or asserts anything about their size.

- **No progress toward Goldbach.** Nothing here bounds $C(N)$, and nothing here is a step toward bounding it.
- **The identities of Section 2 are rearrangements.** Six of the seven are exact and finite, none of those is an estimate, and none makes any quantity smaller. Proposition 13 is the exception: it carries an $O_A(N(\log N)^{-A})$ and uses an asymptotic bound on $|C(N)|$, so it is a consequence of the rearrangements rather than one of them.

1.5 Related work

The present obstruction is methodological rather than a new parity theorem. Bombieri's asymptotic sieve is the classical background for this viewpoint [2]. Two recent works concern the same Huang–Li setting from other directions. Cantarini studies weighted diagonal Möbius-twisted Elliott–Halberstam averages under GRH and a weak Gonek–Hejhal hypothesis, with Sobolev and Hölder–Zygmund weights [1]; those conditional average estimates are different from the support/extraction tradeoff proved here. Moya identifies and repairs the moving-cut omission in the published Huang–Li rearrangement [6]. The weighted switch used below is proved directly in Proposition 2, so the no-go theorem does not depend on the unrepaired moving-cut identity.

2 The design space, exactly

Proposition 4 (the discrepancy is a dilate). *Unconditionally,*

$$A(N; k) = \mu(k) H(N; k), \quad H(N; k) = \sum_{\substack{m < N/k \\ (m, k)=1}} \Lambda(N - mk) \mu(m). \quad (6)$$

Proof. The class $n \equiv N \pmod{k}$ is $k \mid N - n$, so writing $N - n = mk$ turns the sum into $\sum_{m < N/k} \Lambda(N - mk) \mu(mk)$; and $\mu(mk) = \mu(m)\mu(k)$ when $(m, k) = 1$, zero otherwise. $\square$

Identity (6) qualifies the shape of the whole problem. EH_μ is consumed where one Möbius argument *divides* the other, but the per-modulus discrepancy is $\mu(k)$ times a sum in which the two arguments are related by a *difference*, at scale N/k . The two couplings are not two fields; they are the same field seen before and after the switch. It also suggests the source of the difficulty on the μ side: $|A(N; k)| = |H(N; k)|$ is a Möbius–prime correlation of length N/k with no pointwise cancellation supplied by the identity itself. For comparison only, an independent-sign model for $\sum_m \Lambda(N - mk) \varepsilon(mk)$ has square-root-scale cancellation; that model is heuristic and is not used in any proof.

Proposition 5 (the weights are nonnegative). *Substituting (6) into $T_{\log}$ and using $\mu(k)^2 = 1$ on squarefree k ,*

$$T_{\log} = \sum_{\substack{k < K \\ (k, N)=1}} \mu^2(k) (\log k) H(N; k) - C(N) B_{\log}(K), \quad B_{\log}(K) = \sum_{\substack{k < K \\ (k, N)=1}} \frac{\mu(k) \log k}{\varphi(k)}. \quad (7)$$

Proof. Immediate from (6) and $\mu(k)\mu(k) = \mu^2(k)$. $\square$

The signs $\mu(k)$ cancel exactly: they cancel against the $\mu(k)$ that (6) extracts from the progression sum, and what is left is a sum of dilated Möbius–prime correlations with *nonnegative* weights $\mu^2(k) \log k$. Huang–Li discard the $\mu(k)$ by the triangle inequality; on the $w_k = 1$ branch keeping them is what makes the unconditional theorem of [9, Theorem 1] possible; on the $\log k$ branch they are not there to keep. Whatever smallness $T_{\log}$ has must therefore come from the signs of $H(N; k)$ across k . On the finite ranges measured below that cancellation is modest relative to the independent-sign reference. Writing $G = \sum_k (\log k) |H| / |\sum_k (\log k) H|$ for the gain that cancellation across dilations buys, G reads 1.834, 1.804, 2.207, 2.588 and 2.789 over $N = 2 \cdot 10^5$ to $3.2 \cdot 10^6$, where the sums run over 313 to 1485 moduli (`lab_positive_weights.py`).

What independent signs would give is not $\sqrt{\#k}$. Writing $x_k = (\log k) H(N; k)$, the quantity G is $\|x\|_1 / |\sum_k x_k|$, and under independent signs $|\sum_k x_k|$ is of the size of $\|x\|_2$; so the reference for G is $\|x\|_1 / \|x\|_2$. That equals $\sqrt{\#k}$ only when every $|x_k|$ is the same, and Cauchy–Schwarz makes it at most $\sqrt{\#k}$ otherwise. Here the magnitudes are not equal: the same run prints $\|x\|_1 / \|x\|_2$ as 11.96, 14.23, 17.90, 21.43 and 26.21, which is 0.66 to 0.69 of the count-based figure. **A count of moduli overstates the reference by about a half.**

Against the reference that is correct for these magnitudes, the measured G of 1.8 to 2.8 is still smaller by an order of magnitude, so the conclusion is unchanged and only the comparison moves. The step from a signed sum to its absolute counterpart — which is what (13) makes below — costs a factor near two on this range, where independence would have bought between twelve and twenty-six.

Proposition 6 (both ends are the same sum of dilated walls). *For any weight w ,*

$$T_w = \sum_{\substack{k < K \\ (k, N)=1}} \mu^2(k) w_k H(N; k) - B_w C(N). \quad (8)$$

In particular

$$T_1 = \sum_{\substack{k < K \\ (k, N)=1}} \mu^2(k) H(N; k) - B_1 C(N), \quad T_{\log} = \sum_{\substack{k < K \\ (k, N)=1}} \mu^2(k) (\log k) H(N; k) - B_{\log} C(N),$$

and both weights are nonnegative on the range.

Proof. Put (6) into (2). The transform carries $\mu(k)$, so terms with $\mu(k) = 0$ contribute nothing on the left; on the remaining k , $\mu(k) A(N; k) = \mu(k)^2 H(N; k)$. The restriction is therefore carried on the right, and the subtracted mean term assembles into $B_w C(N)$ by (4). $\square$

Note 7 (the restriction is not decoration). Dropping $\mu^2(k)$ from (8) does not weaken it, it falsifies it: $H(N; k)$ does not vanish at squarefull k , while the left-hand side does. The terms so admitted are of the size of T_w itself — at $\theta' = 0.56$ and $N = 2 \cdot 10^5$ they contribute $-0.054316 N$ at $w = 1$ and $-0.287165 N$ at $w = \log$, the latter being 0.656087 of $|T_{\log}|$ in absolute value. The fraction is not stable: over $N = 2, 4, 8 \cdot 10^5$ it reads 0.656087, 0.701640, 0.788480, so it rises with N and the two thirds is the value at the bottom of that range. It stays above $1/2$ throughout, which is the whole of what this note asserts — the restriction is load-bearing and not cosmetic (`audit_squarefull_admitted.py`).

Since $B_1 \ll e^{-c\sqrt{\log K}}$ makes the subtracted mean negligible, the unconditional theorem of [9, Theorem 1] says that the *flat* sum of dilated walls is $\ll_A N (\log N)^{-A}$. On the logarithmic branch, $B_{\log}(K) = -\mathfrak{S}(N) + o(1)$ and the corresponding weighted wall sum is the open object: Proposition 13 shows that an $o(N)$ bound for that signed sum gives $\tilde{r}(N) = \mathfrak{S}(N)N + o(N)$. Thus the visible change between the settled flat branch and this open signed sum is the factor $\log k$ inside the same nonnegative outer weights $\mu^2(k) w_k$.

Measurement 8 (the two ends, measured). Over $N = 2 \cdot 10^5$ to $2.56 \cdot 10^7$ by doubling, at $\theta' = 0.56$, (8) holds to a worst relative error of $1.875 \cdot 10^{-16}$, and

$$\frac{|\sum_k \mu^2(k) H(N; k)|}{|\sum_k \mu^2(k) (\log k) H(N; k)|} = 0.1785, 0.1559, 0.1484, 0.1483, 0.1367, 0.1334, 0.1241, 0.1135,$$

falling on average; at the top $|T_1|/N = 0.01425$ against $|T_{\log}|/N = 0.1245$. Fitted against $\log N$ the flat sum decays as $N^{-0.3505}$ and $|T_{\log}|/N$ as $N^{-0.2658}$: the gap widens. Every fitted exponent in this paper is one arithmetic family's. These N are $2 \cdot 10^5 \cdot 2^j$ with $2 \cdot 10^5 = 2^6 \cdot 5^5$, so each has odd radical 5, and the same doubling carries the -0.3620 below and the residual exponent of Measurement 15; `c02_flatsum_k1.txt` states the field for that reason. What these exponents do across radicals is not measured, and no step of the no-go rests on their values. (The neighbouring -0.2696 printed in the same run is the exponent of $|\sum_k (\log k) H(N; k)|/N$, which is a different sum — the two differ by $C(N)B_{\log}(K)$ — and not a second measurement of the same one. Both fall out of one sieve and one index set, so what separates them is the quantity and not the implementation.) The index set here runs from $k = 1$, as (2) and (8) require; note that $H(N; 1) = C(N)$ exactly, so dropping that one term removes $C(N)$ from the flat sum — which changes the ratios above by up to 11.6% — taken over the $k \geq 1$ column, which is the one those ratios are formed from; against the $k \geq 2$ column it is 10.4%. Those two percentages are not printed by any run: they are the largest relative gap between the `ratio_code` and `ratio_paper` columns of `c02_flatsum_k1.txt`, attained at $N = 4 \cdot 10^5$ where the columns read 0.1740 and 0.1559, and each is that gap divided by one column in turn. The fitted exponent moves to -0.3620 , while leaving $|T_1|$ untouched to 10^{-17} , since the $k = 1$ term of T_1 is $A(N; 1) - C(N)/\varphi(1) = 0$. The two columns are computed side by side, from a sieve built independently of the one `lab_weight_gap.py` uses, in `c02_flatsum_k1.py`; that script's loop is the $k \geq 1$ one, and `lab_weight_gap.py`'s starts at $k = 2$. Identity (6) itself is verified over every squarefree $k < N^{0.56}$ coprime to N at $N = 2 \cdot 10^5, 4 \cdot 10^5, 8 \cdot 10^5$ with worst relative error $2.294 \cdot 10^{-14}, 1.197 \cdot 10^{-13}, 4.591 \cdot 10^{-14}$, and (7) against a direct evaluation of $T_{\log}$ to at most $1.8 \cdot 10^{-16}$ over $N = 2 \cdot 10^5$ to $3.2 \cdot 10^6$ (`lab_weight_gap.py`, `lab_dilate_identity.py`, `lab_positive_weights.py`).

Proposition 9 (the untruncated sum is the count). *Summing over all k , with no truncation and no coprimality restriction,*

$$\sum_{k \geq 1} (\log k) \mu(k) A(N; k) = - \sum_{n < N} \Lambda(n) \mu(N - n) \Lambda(N - n) = \sum_{p < N} \Lambda(N - p) \log p. \quad (9)$$

Thus the untruncated object is the prime-supported weighted sum on the right. It differs from $\tilde{r}(N)$ by $O(N^{1/2} \log^2 N)$, coming from proper prime powers in the second variable.

Proof. Both exchanges are finite sums. Since $A(N; k)$ sums n over $k \mid N - n$, exchanging the order and applying $\sum_{d|u} \mu(d) \log d = -\Lambda(u)$ — Möbius inversion of $\log = \mathbf{1} * \Lambda$ — gives the first equality. The second holds because $\mu(u) \Lambda(u) = -\log p$ at $u = p$ prime and vanishes at every higher prime power. $\square$

Proposition 9 is not a decoration: it says that the untruncated $w = \log$ object is an exact rearrangement of the prime-supported part of $\tilde{r}(N)$, with only the explicit $O(N^{1/2} \log^2 N)$ proper-prime-power discrepancy separating the two. Thus the substantial new difficulty in the switched expression is the truncation at K , not an unquantified algebraic remainder.

Proposition 10 (the count as signed layers). *Cutting the same double sum along the inner variable of (6),*

$$\sum_k \mu^2(k) (\log k) H(N; k) = \sum_m \mu(m) L(N; m), \quad L(N; m) := \sum_{\substack{k < N/m, \\ \mu(k) \neq 0}} (\log k) \Lambda(N - mk), \quad (10)$$

and every $L(N; m)$ is nonnegative. With Proposition 9, this prime-supported weighted pair sum is therefore an alternating sum, signed by μ , of nonnegative layers.

Proof. A finite exchange; the coprimality conditions on the pair (k, m) are symmetric. Non-negativity is that of $\log k$ on $k \geq 1$ and of Λ . The exchange and the nonnegativity are checked numerically in `lab_layer_decomposition.py`, which also measures how much of the total the layers below the truncation carry. $\square$

Proposition 11 (the count over a combined modulus). *Expanding the squarefree condition inside (10) by $\mu^2(k) = \sum_{d^2|k} \mu(d)$ and writing $k = d^2j$,*

$$\sum_{p < N} \Lambda(N - p) \log p = \sum_{\substack{m, d \text{ squarefree} \\ (d, m)=1, md^2 < N}} \mu(m) \mu(d) L_2(N; m, d), \quad (11)$$

where

$$L_2(N; m, d) = \sum_{\substack{j < N/(md^2) \\ (j, m)=1}} \log(d^2j) \Lambda(N - md^2j),$$

and L_2 carries no squarefree condition. It is not quite a single progression: the surviving $(j, m) = 1$ unfolds by Möbius, $j = ej'$ with $e \mid m$, into progressions to moduli md^2e . Only squarefree e survive $\mu(e)$, so $e \mid \text{rad}(m)$ and L_2 is a signed sum of at most $2^{\omega(m)}$ Bombieri–Vinogradov objects with moduli running up to $md^2 \text{rad}(m) = m^2d^2$, since m is squarefree in the sum. What the squarefree condition bought is that none of them carries a μ ; what it did not buy is a single modulus.

Proof. Finite exchange, and $(k, m) = 1$ splits as $(d, m) = (j, m) = 1$. $\square$

Measurement 12 (how large that modulus has to be). Being free of Möbius is a statement about shape, not about level, and the level is the whole question. The truncation below is taken at $md^2 < Q$, while Proposition 11 unfolds that into moduli up to $md^2 \text{rad}(m)$. So the Q^* reported here is a lower bound on the level actually needed — the conservative direction, since what is claimed is that the level must be *large*, and a lower bound that is already large settles that a fortiori. Verified over $N = 2 \cdot 10^5$ to $1.6 \cdot 10^6$ — 166384 to 1331048 pairs — (11) holds to a worst relative error of $4.525 \cdot 10^{-15}$. Truncating it to $q < Q$ and letting $Q^*(N)$ be the least Q past which the deficit stays under εN at the chosen tolerance $\varepsilon = 0.01$,

$$Q^* = 160112, 200848, 482589, 589203, \quad Q^*/\sqrt{N} = 358.0213, 317.5686, 539.5509, 465.8059,$$

so at that tolerance Q^* exceeds $\sqrt{N}$ by two orders at every N tested. Two limits on that reading, with different grounds, and the difference between them matters.

The first is visible in the four numbers themselves: the ratio is not monotone in N : the second entry is below the first and the fourth is below the third, while the third is above both of them. There is no trend in N here to read, at this tolerance, and none is claimed.

The second is not visible here and is not measured here. The tolerance is chosen and not derived, and *no exponent may be fitted to Q^** : the deficit oscillates while decaying, so $Q^*(\varepsilon)$ reads the last excursion above εN rather than a trend, and the ratio has no consistent trend in N across tolerances. The sweep establishing that is not in this packet, so the statement about tolerances is reported from the source and not established here.

What stands is therefore a comparison at one fixed tolerance and no more: expanding the squarefree condition buys the Möbius-free moduli and costs two orders in the size of Q^* at $\varepsilon = 0.01$. No rate is claimed, and this statistic does not measure one (`lab_combined_modulus.py`).

Proposition 13 (the wall cancels out of the count). *Substituting (7) into the identity of [9, Theorem 3], the two occurrences of $C(N)$ carry opposite signs and cancel:*

$$\tilde{r}(N) = \mathfrak{S}(N)N + \sum_{\substack{k < K \\ (k, N)=1}} \mu^2(k)(\log k)H(N; k) - C(N)(B_{\log}(K) + \mathfrak{S}(N)) + O_A\left(\frac{N}{(\log N)^A}\right). \quad (12)$$

Since, for $K = N^{\theta'}$, $B_{\log}(K) + \mathfrak{S}(N) \ll e^{-c_1\sqrt{\log K}} \log K$, the quantity $C(N)$ enters the displayed identity only through a coefficient tending to zero in this regime. Consequently

$$\sum_{\substack{k < K \\ (k, N)=1}} \mu^2(k)(\log k)|H(N; k)| \leq (1 - \varepsilon) \mathfrak{S}(N) N \quad (13)$$

implies

$$\tilde{r}(N) \geq \varepsilon \mathfrak{S}(N)N + o(N),$$

and hence $r_{\mathbb{P}}(N) > 0$ for all sufficiently large even N . Thus (13) is a sufficient condition for binary Goldbach.

Proof. The infinite coefficient and the truncation error are the one input here that (12) does not itself supply. With $F(s) = \prod_{p \nmid N} (1 - \frac{p^{-s}}{p-1})$ one has, after continuation to the origin, $B_{\log}(\infty) = -F'(0)$. Writing $F(s) = G(s)\zeta(s+1)^{-1}$ gives $G(0) = \mathfrak{S}(N)$ and hence $B_{\log}(\infty) = -\mathfrak{S}(N)$. Huang–Li’s Lemma 1, applied to ρ_N and summed by parts against $\log k$ on the tail $k \geq K$, gives the quantitative form actually used: $B_{\log}(K) + \mathfrak{S}(N) \ll e^{-c_1\sqrt{\log K}} \log K = o(1)$ for $K = N^{\theta'}$ with fixed θ' .

Given that, (13), (12) and the trivial bound $|C(N)| \leq \psi(N) \ll N$ give

$$\tilde{r}(N) \geq \varepsilon \mathfrak{S}(N)N + o(N).$$

For even N one has $\mathfrak{S}(N) \geq 2 \prod_{p>2} (1 - (p-1)^{-2}) > 0$, so the right-hand side is $\gg_{\varepsilon} N$. The relation $\tilde{r}(N) = r_{\mathbb{P}}(N) + O(N^{1/2} \log^2 N)$ from the introduction then gives $r_{\mathbb{P}}(N) > 0$ for all sufficiently large even N . $\square$

Note 14 (what the cancellation buys, and what it does not). Condition (13) has the same shape as the constant-factor condition of [9, Proposition 6]: a bound at level $N^{\theta'}$, with no saving in $\log N$. Nothing has moved off the level axis. What has changed is the threshold constant, from $\mathfrak{S}(N)(1 - \mathfrak{A}(N))$ to $\mathfrak{S}(N)$. The change removes the collapsing factor $1 - \mathfrak{A}(N)$ and hence the strongest arithmetic sensitivity tied to that factor; $\mathfrak{S}(N)$ itself still depends on the prime divisors of N and is not claimed to be constant. This is offered as a weaker sufficient condition, proved from two identities. It is not offered as something computation confirms: at accessible N the unspecified O_A term of (12) is numerically the very quantity (13) must bound.

Measurement 15 (the two thresholds, compared). At $\theta' = 0.56$ over $N = 2 \cdot 10^5$ to $3.2 \cdot 10^6$, with $B_H(N) = \sum_{k < K} \mu^2(k)(\log k)|H(N; k)|$, the new ratio $B_H(N)/(\mathfrak{S}(N)N)$ reads 0.4578, 0.4064, 0.4079, 0.3769, 0.3338 — under 1 throughout, and falling overall though not at every step, the third entry sitting above the second — against 2.1519, 1.9105, 1.9175, 1.7720, 1.5691 for the same numerator against the old threshold, $B_H(N)/(\mathfrak{S}(N)(1 - \mathfrak{A}(N))N)$, which is above 1 throughout. *The numerator is held fixed across the two lists; only the threshold constant moves, which is the whole of the comparison.* (For the numerator of [9, Proposition 6], which keeps the subtracted mean inside the absolute value and so is a different sum, the old ratio reads 2.1591, 1.9747, 1.9500, 1.7483, 1.5798 — the same conclusion, a different quantity.) Across a seven-point family of arithmetically diverse N the new ratio spans a factor 2.00, where the same family taken against the old threshold with *that*

other numerator spans 23.25. The two spans are not formed from the same sum, and the run prints them under column names that say so; what the spans compare is the two thresholds, and the arithmetic sensitivity they differ by belongs to $1 - \mathfrak{A}(N)$, which is the only factor between them. The residual coupling is negligible at these N : $|B_{\log} + \mathfrak{S}|/\mathfrak{S}$ stays under 0.0383 and $|C(N)(B_{\log} + \mathfrak{S})|/N$ never exceeds $3.315 \cdot 10^{-4}$, four orders below $\mathfrak{S}(N)$. The residual of (12) decays as $N^{-0.2794}$ with correlation -0.99930 and is extrapolated by the fitted power law to reach 2% of N near $N = 10^{10.16}$ (`lab_direct_route.py`, `lab_direct_identity.py`; the two lists held to one numerator, and the third quantity in the parenthesis, are recomputed side by side in `a1_bh_vs_b.py` — `lab_direct_route.py` prints the parenthetical list and the first of the two above it; only the second, 2.1519, $\dots$, is exclusive to `a1_bh_vs_b.py`).

3 The no-go

3.1 The two opposing constraints

Lemma 16 (extraction).

$$B_w = \sum_{\substack{d < K \\ (d, N)=1}} b_d \frac{\mu(d)}{\varphi(d)} \rho_{dN} \left(\frac{K}{d} \right), \quad \rho_n(x) = \sum_{j < x, (j, n)=1} \frac{\mu(j)}{\varphi(j)},$$

and $\rho_n(x) \ll e^{-c_1 \sqrt{\log x}}$ whenever $\log n \ll \log x$.

Proof. Write $k = dj$; for squarefree k one has $(d, j) = 1$, $\mu(k) = \mu(d)\mu(j)$ and $\varphi(k) = \varphi(d)\varphi(j)$. The bound on ρ is Huang–Li’s Lemma 1, a form of Goldston–Yıldırım [3], and the hypothesis $\log n \ll \log x$ is theirs. $\square$

Note 17 (the hypothesis is load-bearing, and it is met where it is used). Without $\log n \ll \log x$ the bound is false: at $\lfloor K/d_0 \rfloor = 1$ the run behind Measurement 25 prints $|B_w|\varphi(d_0) = 1.000000$ over all 5130 admissible d_0 (`audit_extraction_tradeoff.py`; that row is not among the two the measurement reproduces), no saving at all, and that is exactly the corner where $\log n \asymp \log N$ while $\log x = O(1)$. Theorem 20 is unaffected: it maximises only over $d \leq N^{\theta' - 1/2 - \delta}$, where $x = K/d \geq N^{1/2 + \delta}$ and $n = dN \leq N^{3/2}$, so $\log n \asymp \log x$ holds throughout the range it uses. The same applies to Theorem 24.

Lemma 18 (weighted maximal Bombieri–Vinogradov). *For every $A, B > 0$ there is $C = C(A, B)$ such that, with*

$$\mathcal{E}_q(N) := \max_{y \leq N} \max_{(a, q)=1} \left| \sum_{\substack{n \leq y \\ n \equiv a \pmod{q}}} \Lambda(n) - \frac{y}{\varphi(q)} \right|,$$

one has

$$\sum_{q \leq Q} \tau_3(q)^B \mathcal{E}_q(N) \ll_{A, B} \frac{N}{(\log N)^A} \quad (Q \leq N^{1/2} (\log N)^{-C}).$$

Proof. This is a standard weighted consequence of the maximal Bombieri–Vinogradov theorem; see, for example, [8, §20.2]. By Cauchy–Schwarz,

$$\sum_{q \leq Q} \tau_3(q)^B \mathcal{E}_q \leq \left(\sum_{q \leq Q} \tau_3(q)^{2B} \mathcal{E}_q \right)^{1/2} \left(\sum_{q \leq Q} \mathcal{E}_q \right)^{1/2}.$$

The trivial estimate $\mathcal{E}_q \ll (N/\varphi(q)) \log N$ and the standard mean-value bounds for fixed powers of the divisor function give

$$\sum_{q \leq Q} \tau_3(q)^{2B} \mathcal{E}_q \ll_B N (\log N)^{C_1(B)}.$$

The maximal Bombieri–Vinogradov theorem gives, for every $D > 0$, a constant $C_2(D)$ such that

$$\sum_{q \leq Q} \mathcal{E}_q \ll_D \frac{N}{(\log N)^D} \quad (Q \leq N^{1/2} (\log N)^{-C_2(D)}).$$

Taking $D = 2A + C_1(B) + 2$ and enlarging the logarithmic modulus loss if necessary proves the lemma. No distribution theorem stronger than the usual Bombieri–Vinogradov theorem is used. $\square$

Lemma 19 (termwise BV accessibility). *Expanding $w_k = \sum_{d|k} b_d$ in the residual of Proposition 2 gives $\mathcal{R}_w = \sum_d b_d \mathcal{R}^{(d)}$. For every fixed $\delta > 0$ and every $A > 0$,*

$$\mathcal{R}_w \ll_{A, \delta, \theta'} \|b\|_1 \frac{N}{(\log N)^A}$$

whenever b is supported on $d \leq N^{\theta' - 1/2 - \delta}$.

Proof. If d is not squarefree or $(d, N) > 1$, then $\mathcal{R}^{(d)} = 0$. Otherwise write $k = d\ell$, put $K_d := K/d$, and then put $r = md$. Because the residual already has $(k, mN) = 1$, one has $(m, d) = 1$ and $\mu(m) = \mu(d)\mu(r)$. Hence

$$\mathcal{R}^{(d)} = \mu(d) \sum_{\substack{r < N/K_d \\ d|r}} \mu(r) \sum_{\substack{\ell \geq K_d, r\ell < N \\ (\ell, rN)=1}} \mu^2(\ell) \Lambda(N - r\ell).$$

The support hypothesis gives $K_d \geq N^{1/2+\delta}$ and $R_d := N/K_d \leq N^{1/2-\delta}$.

We first dispose of the non-coprime prime classes. If $(\ell, N) > 1$ and $\Lambda(N - r\ell) \neq 0$, choose $p \mid (\ell, N)$. Then $p \mid N - r\ell$; hence $N - r\ell = p^\nu$ for some $p \mid N$. There are $\ll (\log N)^2$ such prime powers, and for each one the pairs (r, ℓ) with $r\ell = N - p^\nu$ number at most $\tau(N - p^\nu)$. Thus these terms contribute $O(N^{o(1)})$, uniformly in d . We may therefore replace $(\ell, rN) = 1$ by $(\ell, r) = 1$ at this cost.

Now insert

$$\mu^2(\ell) = \sum_{a^2|\ell} \mu(a), \quad \mathbf{1}_{(\ell, r)=1} = \sum_{e|(\ell, r)} \mu(e),$$

and truncate both auxiliary variables at $Y := \exp(\sqrt{\log N})$. The discarded $a > Y$ terms are bounded, after dropping the restriction $d \mid r$, by

$$\ll \log N \sum_{r < R_d} \sum_{Y < a \leq \sqrt{N/r}} \left(\frac{N}{ra^2} + 1 \right) \ll \frac{N(\log N)^2}{Y} + \sqrt{NR_d} \log N.$$

For $e > Y$ (where $e \mid r$ and r is squarefree), the same positive progression count gives

$$\ll \frac{N(\log N)^3 \log \log N}{Y} + R_d Y (\log N)^2.$$

Both bounds are $\ll_A N(\log N)^{-A}$ for every fixed A . In the remaining range the prime variable $n = N - r\ell$ lies in a residue class $n \equiv N \pmod{q}$ with

$$q = r \operatorname{lcm}(a^2, e) \leq R_d Y^3 \leq N^{1/2-\delta/2}$$

for all sufficiently large N .

If $(q, N) > 1$ and $\Lambda(n) \neq 0$, then any prime $p \mid (q, N)$ also divides n , so again $n = p^j$ with $p \mid N$. The number of such n is $\ll (\log N)^2$. For a fixed n , a triple (r, a, e) contributing to such a class satisfies $r \operatorname{lcm}(a^2, e) \mid N - n$, hence there are at most $\tau(N - n)^3 = N^{o(1)}$ possibilities. Thus the non-coprime moduli also contribute only $O(N^{o(1)})$, uniformly in d .

For the remaining classes, a modulus q arises from at most $\sum_{r|q} \tau(q/r)^2 \leq \tau_3(q)^3$ triples (r, a, e) . Lemma 18, with $B = 3$, therefore gives the total progression error $O_A(N(\log N)^{-A})$, uniformly in d .

It remains to bound the progression main term. For squarefree r with $(r, N) = 1$ define the full density

$$c(r) := \sum_{a \geq 1} \sum_{e|r} \frac{\mu(a)\mu(e)\mathbf{1}_{(a,N)=1}}{\varphi(r \operatorname{lcm}(a^2, e))}.$$

This Euler product is elementary. If $p \nmid r$, its local factor is $1 - 1/\varphi(p^2) = 1 - 1/(p(p-1))$ for $p \nmid N$, and is 1 for $p \mid N$. If $p \mid r$ (hence $p \nmid N$), the four choices $(v_p(a), v_p(e)) \in \{0, 1\}^2$ contribute

$$\frac{1}{p-1} - \frac{1}{p(p-1)} - \frac{1}{p^2(p-1)} + \frac{1}{p^2(p-1)} = \frac{1}{p}.$$

Consequently

$$c(r) = \mathfrak{A}(N) \frac{\lambda(r)}{r}, \quad \lambda(r) = \prod_{p|r} \left(1 - \frac{1}{p(p-1)}\right)^{-1}.$$

The truncated density differs from $c(r)$ by

$$O\left(\frac{\tau(r)}{\varphi(r)Y}\right) + O\left(\frac{\tau(r) \log \log(3R_d)}{\varphi(r)Y}\right).$$

Indeed, for $a > Y$ use that $r \operatorname{lcm}(a^2, e)$ is a multiple of ra^2 ; for $e > Y$ sum first over a and compare prime by prime with $\varphi(re)$. Since $r = md$, $(m, d) = 1$, and d is odd squarefree, $\tau(d)/\varphi(d) \leq 1$, while $\sum_{m < M} \tau(m)/\varphi(m) \ll (\log M)^2$. After multiplication by $N - mK \leq N$, the total density-truncation error is therefore $\ll N(\log N)^3 \log \log N/Y$, below every fixed log-power target.

Since $r = md$ and $(m, dN) = 1$, multiplicativity gives the remaining main term

$$\mathfrak{A}(N) \frac{\lambda(d)}{d} \sum_{\substack{m < N/K \\ (m, dN)=1}} \frac{\mu(m)\lambda(m)}{m} (N - mK).$$

To see the required cancellation without importing another weighted switch, set

$$G_{d,N}(x) := \sum_{\substack{m \leq x \\ (m, dN)=1}} \frac{\mu(m)\lambda(m)}{m}.$$

The Dirichlet series of $\mu(m)\lambda(m)\mathbf{1}_{(m, dN)=1}$ factors as $\zeta(s)^{-1}H_{d,N}(s)$. Writing its coefficients as $\mu * h_{d,N}$, the local factors satisfy $h_{d,N}(p^j) = 1$ for $p \mid dN$ and $h_{d,N}(p^j) = 1 - \lambda(p) = O(p^{-2})$ otherwise. The classical zero-free-region estimate $\sum_{a \leq y} \mu(a)/a \ll e^{-c\sqrt{\log y}}$; see, for example, [7]. It then gives, after splitting the convolution at $b = \sqrt{x}$,

$$G_{d,N}(x) \ll \frac{dN}{\varphi(dN)} e^{-c\sqrt{\log x}} + x^{-1/4} e^{C\sqrt{\log(dN)}}.$$

Here $dN \leq N^{3/2}$ and $M = N/K = N^{1-\theta'}$ is a fixed power of N . Put $x_0 = \exp(4C\sqrt{\log(dN)}) = N^{o(1)}$ uniformly in d . On $[1, x_0]$ use $|G_{d,N}(x)| \ll \log x$; on $[x_0, M]$ use the last display. Abel summation against $N - mK$ yields

$$Kx_0 \log x_0 + N \log \log N e^{-c'\sqrt{\log M}} + KM^{3/4} e^{C\sqrt{\log(dN)}} \ll_A \frac{N}{(\log N)^A}.$$

The bounded prefactor $\lambda(d)/d$ does not change the estimate. Combining the progression error, the main term, the truncation tails and the degenerate classes gives $\mathcal{R}^{(d)} \ll_A N(\log N)^{-A}$ uniformly in d . Summing with weights $|b_d|$ proves the lemma. $\square$

The two lemmas pull in opposite directions. Lemma 16 says B_w is controlled by the part of b sitting at the truncation point K : mass at $d \leq K^{1-\varepsilon}$ is damped by $e^{-c\sqrt{\varepsilon \log K}}$. Lemma 19 says the residual is reachable by this termwise use of Bombieri–Vinogradov when b lives below $N^{\theta'-1/2}$; it does not say that a b living higher puts the residual out of reach. Within this sufficient support class, the extraction coefficient is damped across a factor $N^{1/2}$.

3.2 The theorem

Theorem 20 (loss in the termwise BV/triangle-inequality extraction). *Fix $\theta' \in (1/2, 1)$ and $0 < \delta < \theta' - \frac{1}{2}$. Let $w = w_N$ be a real weight with transform $b = \mu * w$, supported in*

$$1 \leq d \leq D := N^{\theta'-1/2-\delta}.$$

This support condition is a sufficient condition for the termwise Bombieri–Vinogradov estimate of Lemma 19; it is not asserted to be necessary. If $B_w \neq 0$, then

$$\frac{\|b\|_1}{|B_w|} \gg \exp\left(c_1 \sqrt{\left(\frac{1}{2} + \delta\right) \log N}\right). \quad (14)$$

If, in addition, one grants the homogeneous bound

$$|T_w| \ll_A \|b\|_1 \frac{N}{(\log N)^A} \quad \text{for every fixed } A > 0,$$

then the triangle-inequality evaluation of (5) gives only

$$|C(N)| \leq \frac{\|b\|_1}{|B_w|} \cdot O_A\left(\frac{N}{(\log N)^A}\right).$$

Because the prefactor in this particular majorant is larger than every fixed power of $\log N$, these inequalities, with any fixed choice of A , do not by themselves imply a fixed log-power saving for $C(N)$.

The scope is part of the conclusion. Weights outside the support hypothesis are not covered. Nor are evaluations that exploit cancellation among the b_d , or cancellation among the complete term, $\mathcal{R}_w$ and T_w in (5). The theorem closes the two triangle-inequality steps just displayed, not every possible use of the switch identity.

Proof. By Lemma 16 and $\varphi(d) \geq 1$,

$$|B_w| \leq \sum_{d \leq D} \frac{|b_d|}{\varphi(d)} \left| \rho_{dN}\left(\frac{K}{d}\right) \right| \leq \|b\|_1 \max_{d \leq D} \left| \rho_{dN}\left(\frac{K}{d}\right) \right|.$$

Here $K/D = N^{1/2+\delta}$, while $dN \leq N^{\theta'+1/2-\delta}$, so the hypothesis $\log(dN) \ll \log(K/d)$ of Lemma 16 holds uniformly. Thus

$$|B_w| \ll \|b\|_1 e^{-c_1 \sqrt{\left(\frac{1}{2} + \delta\right) \log N}},$$

which is (14).

The weighted switch (5) is homogeneous in b . Its complete part satisfies

$$\left| \sum_{u < N} \Lambda(N-u) \mu^2(u) b_u \right| \leq (\log N) \|b\|_1,$$

the switch error is $O(\|b\|_1 N^{o(1)})$ by Proposition 2, and the residual is

$$|\mathcal{R}_w| \ll_A \|b\|_1 \frac{N}{(\log N)^A}$$

by Lemma 19. Add the granted bound on T_w and divide by $|B_w|$. Since $N^{o(1)} \ll_A N(\log N)^{-A}$ for every fixed A after weakening the exponent if necessary, the displayed upper bound for $|C(N)|$ follows.

Finally, (14) is a lower bound on the coefficient appearing in this *particular majorant*; it is not a lower bound for $C(N)$. Consequently the displayed chain of termwise estimates, for any fixed choice of A , cannot on its own certify a fixed log-power saving. This is the limited methodological obstruction asserted here; it does not exclude a sharper estimate using cancellation discarded by the triangle inequality. $\square$

Note 21 (why granting T_w is the right form). Without the granted hypothesis the theorem would be vacuous rather than strong: (5) could then be satisfied by T_w alone being large, and it says nothing about $C(N)$ at all. Granting it isolates what the design space actually fails at. The failure is not a shortage of input — the input is handed over — but the arithmetic of two thresholds that lie a factor $N^{1/2}$ apart. Theorem 24 shows that the same support-based loss persists throughout its nonempty fixed-level range $1 - \theta' < \theta_E < 1$.

The loss factor is $\exp(c_1 \sqrt{(\frac{1}{2} + \delta) \log N})$, and the $1/2$ records the square-root separation between the truncation scale $N^{\theta'}$ governing the extraction coefficient and the lower support scale $N^{\theta'-1/2}$ sufficient for this termwise Bombieri–Vinogradov control of the residual. The two known weights exhibit the two endpoints of this space; Theorem 20 shows that on the low-support class admitted by this termwise argument the extraction coefficient is exponentially damped relative to $\|b\|_1$.

Note 22 (which branch the no-go covers). The two endpoints are not on the same side of Theorem 20’s support hypothesis, and the asymmetry is worth stating because it runs opposite to the way the two branches are usually named. That hypothesis asks the transform $b = \mu * w$ to be supported in $[1, N^{\theta'-1/2-\delta}]$, and

$$w_k = 1 \implies b = \mu * 1 = \delta, \quad w_k = \log k \implies b = \mu * \log = \Lambda,$$

by Möbius inversion. The delta is inside the hypothesis for any positive bound; the support of Λ runs to the truncation point, measured at 1369 and 2971 against bounds 2.2 and 2.4 at $\theta' = 0.56$. Those bounds are $N^{\theta'-1/2}$, the $\delta = 0$ endpoint, which is the largest the hypothesis ever allows; any $\delta > 0$ makes them smaller and the gap wider, so the comparison is stated in the direction least favourable to it. So Theorem 20 covers the *flat* branch and not the log one — the flat branch being the one [9, Theorem 1] settles unconditionally, and the log branch the one left open. That is consistent rather than circular: what fails here is the displayed *termwise triangle-inequality* extraction of $C(N)$ on the stated support class, while [9, Theorem 1] does not extract $C(N)$; it shows the weighting is small, which is a different thing.

Below the threshold the statement is empty rather than false. For $\theta' < 1/2$ the support bound $N^{\theta'-1/2-\delta}$ is under 1 — the run prints 0.5 and 0.4 at $\theta' = 0.44$ for $\delta = 0$, which is the largest the bound can be — so no weight meets the hypothesis at all, the delta included. That Theorem 20 says nothing there is not a gap in it: it is the same Bombieri–Vinogradov reach that governs $\theta' > 1/2$ throughout, appearing here in the theorem’s own hypothesis (`audit_nogo_scope.py`).

Note 23 (relation to Bombieri’s asymptotic sieve). The genre of Theorem 20 is classical. Bombieri’s asymptotic sieve [2] is a central reference for the parity obstruction in sieve theory, and it motivates looking for a quantitative obstruction of this shape. Theorem 20 makes only the narrower statement proved here: within the termwise triangle-inequality evaluation, the loss is quantified by the separation between the support allowed by the progression estimate and the scale at which the extraction coefficient ceases to be exponentially damped. The same power-scale loss persists throughout the nonempty range $1 - \theta' < \theta_E < 1$ of Theorem 24, rather than only at Bombieri–Vinogradov level. The reason the statement is worth making is narrow but real: without it, the $\log k$ weight looks like one member of a family of candidate consumptions, and the identity of [9, Theorem 3] then looks like an accident of one choice of weight rather than a property of the route.

3.3 The support-based loss under higher distribution

The obvious objection to Theorem 20 is that Bombieri–Vinogradov is not the only available input: Huang–Li’s Theorem 1 itself *assumes* EH for Λ at level N^θ . Raising the assumed level does not remove the same support-based loss at any fixed level below 1, as the next theorem states.

For the next statement, $EH(N^{\theta_E})$ means the maximal level-of-distribution estimate: for every $A > 0$,

$$\sum_{q \leq N^{\theta_E}} \max_{y \leq N} \max_{(a,q)=1} \left| \sum_{\substack{n \leq y \\ n \equiv a \pmod{q}}} \Lambda(n) - \frac{y}{\varphi(q)} \right| \ll_A \frac{N}{(\log N)^A}.$$

As in Lemma 18, this implies the corresponding fixed divisor-weight version after sacrificing an arbitrarily large power of $\log N$.

Theorem 24 (the same support argument under higher distribution). *Suppose Λ satisfies a level-of-distribution estimate $EH(N^{\theta_E})$ for some fixed $\theta_E \in (0, 1)$. Fix $\eta > 0$ with $\theta_E - (1 - \theta') - \eta > 0$. The same termwise progression argument controls the residual whenever*

$$b_d = 0 \quad (d > N^{\theta_E - (1 - \theta') - \eta}).$$

For weights in that support class with $B_w \neq 0$,

$$\frac{\|b\|_1}{|B_w|} \gg \exp\left(c_1 \sqrt{(1 - \theta_E + \eta) \log N}\right),$$

and hence, a fortiori, by $\exp(c_1 \sqrt{(1 - \theta_E) \log N})$. Thus this support-based triangle-inequality evaluation retains a loss larger than every fixed power of $\log N$ for each fixed $\theta_E < 1$. No necessity statement for the support condition is claimed.

Proof. With $k = d\ell$ the short variable has size $m < N^{1 - \theta'}$, while the auxiliary factors in the progression modulus are subpolynomial (and may be taken polylogarithmic for any prescribed log-power error). The fixed power slack $N^{-\eta}$ therefore puts all resulting moduli inside N^{θ_E} for large N . This gives the residual estimate by the same argument as Lemma 19. For $D = N^{\theta_E - (1 - \theta') - \eta}$,

$$\frac{K}{D} = N^{1 - \theta_E + \eta}.$$

Lemma 16, exactly as in Theorem 20, gives the displayed ratio. $\square$

For every fixed $\theta_E > 1 - \theta'$ one may choose such an η , so the loss persists at every fixed level below 1 supplied by the usual Elliott–Halberstam conjecture. At the formal power-scale endpoint $\theta_E = 1$ the factor $N^{1 - \theta_E}$ disappears, but the theorem neither assumes nor analyses that endpoint; auxiliary factors still require slack. Nothing here rules out a different evaluation of the same residual that exploits cancellation across the transform.

Measurement 25 (the load-bearing lemma, checked as an identity). Lemma 16 carries the theorem, so it is checked directly at $N = 99\,999\,998$, $\theta' = 0.56$, $K = \lfloor N^{\theta'} \rfloor = 30199$. For the single-divisor weights $w_k = [d_0 \mid k]$, i.e. $b = \delta_{d_0}$, the lemma is an identity and is verified as one: over all 10 262 squarefree $d_0 < K$ coprime to N , the brute-force B_w over $k < K$ and the factorised $\mu(d_0)\varphi(d_0)^{-1}\rho_{d_0N}(K/d_0)$ agree to $6.505 \cdot 10^{-18}$, the worst case being $d_0 = 33$.

The finite sweep illustrates the tradeoff in the lemma. Values of $|B_w|\varphi(d_0)$ become large when K/d_0 is small, while they are strongly damped on the Bombieri–Vinogradov support range used by Theorem 20: for these parameters $N^{\theta'-1/2} = 3$, and over $d_0 \leq 3$ the largest value attained is 0.004145. The value is not a function of K/d_0 alone — ρ_{d_0N} carries the condition $(j, d_0N) = 1$, so it depends on which small primes divide d_0 . At $\lfloor K/d_0 \rfloor = 7$ the 184 admissible d_0 produce exactly four values, split by $\gcd(d_0, 15)$:

$\gcd(d_0, 15)$	1	3	5	15
$ B_w \varphi(d_0)$	0.250000	0.750000	0.500000	1.000000

and at $\lfloor K/d_0 \rfloor = 13$ the 55 admissible d_0 produce nine values spanning 0.066667 to 0.816667, the largest at the d_0 divisible by 15 and by no other prime below 14, where $j = 3$ and $j = 5$ drop out of ρ_{d_0N} and what is left is $1 - \frac{1}{10} - \frac{1}{12}$ (`audit_extraction_tradeoff.py`). The list stood here as six values to 0.566667 until the run behind it was found to be printing only the first six of its own column, without saying so; the three it dropped — 0.650000, 0.666667, 0.816667 — were the three largest. So the decay in $\lfloor K/d_0 \rfloor$ is slower than this measurement was read as saying — which does not touch Theorem 20, whose range is $d_0 \leq 3$, i.e. $\lfloor K/d_0 \rfloor \geq 10066$, where the maximum is 0.004145.

4 Smooth weights: $\mu * \log^D = \Lambda_D$

Theorem 20 closes the stated termwise Bombieri–Vinogradov/triangle-inequality evaluation when the transform $b = \mu * w$ is supported sufficiently low. That support hypothesis excludes the most natural family of all, $w_k = f(\log k)$ with f a polynomial, whose transform is spread over every scale. For those the structure is explicit:

$$b = \mu * \log^D = \Lambda_D,$$

the generalised von Mangoldt function. Λ_D vanishes on integers with more than D prime factors, and on a squarefree $u = p_1 \cdots p_r$ with $r \leq D$ it is the r -th finite difference of x^D at the points $\log p_i$; in particular $\Lambda_D(u) = D! \log p_1 \cdots \log p_D$ when $r = D$. Hence for $f = x^D$ the complete part splits by the number of prime factors,

$$\text{CP}_D(N) = \sum_{r=1}^D \sum_{\substack{u < N \\ \text{squarefree} \\ \omega(u)=r}} \Lambda(N-u) \Lambda_D(u), \quad (15)$$

the $r = 1$ piece being a Goldbach-type sum and the $r \geq 2$ pieces Chen-type sums (prime plus r -almost-prime).

Proposition 26 (polynomial weights). *Let $w_k = f(\log k)$ with $f = \sum_{j \leq D} c_j x^j$ real.*

- (i) *If $D = 0$ then $B_w \ll e^{-c\sqrt{\log K}}$ (Huang–Li’s Lemma 1): no extraction.*
- (ii) *If f is a single monomial x^D , $D \geq 1$, then every term of CP_D is nonnegative, since $\Lambda \geq 0$ and $\Lambda_D \geq 0$; so $\text{CP}_D \geq 0$ and no cancellation is available inside it. No lower bound of the required size is obtained here. At $D = 1$ the complete part is*

$$\text{CP}_1(N) = \sum_{p < N} \Lambda(N-p) \log p,$$

which differs from the prime–prime weighted Goldbach count only by proper–prime–power terms of size $O(N^{1/2} \log^2 N)$. A uniform lower bound $\text{CP}_1(N) \gg N$ would therefore imply Goldbach for all large even N , while a full asymptotic is of Hardy–Littlewood type. Neither is known here. Nonnegativity alone does not show that CP_D is large, nor does it show that it is not $o(N)$.

(iii) To obtain cancellation by combining different monomials one would need coefficients of opposite sign. Determining a tuning from leading terms would require information about the prime-supported moments

$$\sum_{p < N} \Lambda(N - p)(\log p)^j,$$

including the $j = 1$ case above. Those asymptotics are not supplied by the present method. Thus the most direct leading-term tuning is circular with respect to the prime-pair information one hoped to avoid; this does not exclude every possible polynomial mechanism.

Consequently this section does not produce a polynomial weight whose complete part is both classically controlled and useful for extraction. It does not prove that no such polynomial weight exists.

Proof. (i) is Huang–Li’s Lemma 1 applied to $w \equiv 1$. For (ii), $\Lambda_D \geq 0$ on the squarefree integers with $\omega \leq D$ because for $u = p_1 \cdots p_r$ the value $\Lambda_D(u)$ is the r -fold mixed difference of $x \mapsto x^D$ at $\log p_1 < \cdots < \log p_r$, and an r -fold difference of a function whose r -th derivative $D(D-1)\cdots(D-r+1)x^{D-r}$ is nonnegative is itself nonnegative; convexity alone would give this only for $r \leq 2$. Every summand of (15) is then a product of nonnegative factors, so no cancellation is available and $\text{CP}_D \geq 0$. What (ii) establishes is only the absence of internal sign cancellation for a monomial complete part. No lower bound of the required order is claimed: the $r = 1$ piece is a prime-plus-prime-power count, and at $D = 1$ it is the whole of CP_1 . After subtracting the $O(N^{1/2} \log^2 N)$ proper-prime-power contribution, a linear lower bound would imply a genuine prime–prime representation. For (iii), the direct leading-term tuning would require asymptotics for the same prime-supported moments, so that particular calibration uses information not provided here. $\square$

Note 27 (the canonical tuning moves the wrong way). Since $b = \mu * w$ we have $w = \mathbf{1} * b$ and hence $B(s) = W(s)/\zeta(s)$. Put $t = s - 1$. For $f(x) = x^2 + cx$, $W(s) = \zeta''(s) - c\zeta'(s)$, and the Laurent expansions at $s = 1$ give

$$B(s) = \frac{2}{t^2} + \frac{c - 2\gamma}{t} + O(1).$$

Thus the coefficient of the double pole is independent of c : a quadratic tuning can change the simple-pole coefficient but cannot remove the leading pole. Equivalently, in the formal Perron residue the main terms are

$$2x \log x + (c - 2 - 2\gamma)x,$$

so the canonical choice $c = 2 + 2\gamma$ cancels the x term, not the $x \log x$ term. This observation concerns the mean of b ; the complete part depends instead on its correlation with $\Lambda(N - \cdot)$.

Which way it moves is measured, and the title is not rhetorical. At $N = 10^6$ the untuned CP is $39.8793 N$; the canonical tuning $x^2 + (2 + 2\gamma)x$ raises it to $45.4201 N$, a rise of 13.89 per cent, while $x^2 - 2\gamma x$ lowers it to $37.8516 N$, by 5.08 per cent. The same ordering holds at $4 \cdot 10^6$ and $1.6 \cdot 10^7$ (`audit_polyweight.py`). Neither tuning buys smallness, and the canonical one is the worse of the two: killing the mean of b is not what is wanted. What must be small is b ’s correlation with $\Lambda(N - \cdot)$, not its mean.

Measurement 28 (the two pieces are the same order). Nonnegativity, and not a separation of scales, is what establishes the limited statement in (ii). At $N = 10^6, 4 \cdot 10^6, 1.6 \cdot 10^7$ the two pieces of CP_2 are the same order:

	N	10^6	$4 \cdot 10^6$	$1.6 \cdot 10^7$
$r = 1 : \sum_p \Lambda(N-p) \log^2 p / N$		22.5145	25.0438	27.4579
$r = 2 : 2 \sum_{pq} \Lambda(N-pq) \log p \log q / N$		17.3648	19.7926	22.2512
ratio r_2/r_1		0.7713	0.7903	0.8104
$\text{CP}_2/(N \log N)$		2.8866	2.9494	2.9967

The ratio drifts toward 1, not toward 0 or ∞ . Calibration: the $D = 1$ column reproduces the Goldbach sum $\sum_p \Lambda(N-p) \log p = 1.7565N, 1.7633N, 1.7614N$ against $\mathfrak{S}(N) = 1.76043$, which is the same at all three N because N has the same prime support $\{2, 5\}$ in each case (`audit_polyweight.py`).

5 Both standard pairings have zero margin on $C(N)$

Since the preceding no-go concerns only the explicit termwise Bombieri–Vinogradov support class, and the polynomial discussion is weaker still, it is useful to record the position of two global Hölder pairings for the other classical representation. Write

$$C(N) = \int_0^1 S_\Lambda(\alpha) S_\mu(\alpha) e(-N\alpha) d\alpha,$$

where

$$S_\Lambda(\alpha) = \sum_{n \leq N} \Lambda(n) e(n\alpha), \quad S_\mu(\alpha) = \sum_{m \leq N} \mu(m) e(m\alpha).$$

Proposition 29 (zero margin). *Two natural global Hölder pairings for this integral — $(2, 2)$ and $(\infty, 1)$ — lie at or above the trivial bound $|C(N)| \leq \psi(N) \sim N$:*

(i) $\|S_\Lambda\|_2 \|S_\mu\|_2 = (\sum_{n \leq N} \Lambda(n)^2)^{1/2} (\sum_{m \leq N} \mu^2(m))^{1/2} \sim (6/\pi^2)^{1/2} N (\log N)^{1/2}$, exceeding the trivial bound by a factor $\asymp (\log N)^{1/2}$, which grows;

(ii) any bound of the shape $\sup_\alpha |S_\mu(\alpha)| \cdot \|S_\Lambda\|_1$ is at least $\|S_\mu\|_2 \|S_\Lambda\|_1 \gg N^{1/2} \cdot N^{1/2} = N$.

Proof. (i) is Cauchy–Schwarz followed by $\sum_{n \leq N} \Lambda(n)^2 \sim N \log N$ and $\sum_{m \leq N} \mu^2(m) \sim 6N/\pi^2$. For (ii), the two factors are bounded below for different reasons. Parseval gives $\sup_\alpha |S_\mu| \geq \|S_\mu\|_2 = (6N/\pi^2)^{1/2}$, an identity-level constraint: no improvement in Möbius exponential-sum technology can lower $\sup_\alpha |S_\mu|$ below $(6/\pi^2)^{1/2} N^{1/2}$. For the second factor Parseval runs the *wrong way*: it gives $\|S_\Lambda\|_1 \leq \|S_\Lambda\|_2 \ll (N \log N)^{1/2}$, an upper bound, while the trivial lower bound $\|S_\Lambda\|_1 \geq |\int_0^1 S_\Lambda(\alpha) e(-n\alpha) d\alpha| = \Lambda(n)$ reaches only $\log N$. The bound $\|S_\Lambda\|_1 \gg N^{1/2}$ that (ii) uses is a theorem of Vaughan [10]. $\square$

The classical uniform bound $S_\mu(\alpha) \ll_A N (\log N)^{-A}$, due to Davenport and quoted here as standard rather than proved, is therefore useless here: it is a saving over N , whereas the pairing needs a bound at the scale $N^{1/2}$, which Parseval forbids. For the global $(\infty, 1)$ pairing, this is a zero-margin obstruction: improving Davenport’s uniform bound while retaining the same global pairing cannot cross the Parseval floor. A major/minor-arc decomposition with bilinear treatment is a different mechanism and is not ruled out here.

Measurement 30 (the constants). Evaluated by FFT on a $4N$ -point grid at $N = 2^{14}, \dots, 2^{20}$. The table is a sampled diagnostic: the continuous norms in Proposition 29 are the theorem-level

objects and no proof uses the numerical rows below.

N	2^{14}	2^{16}	2^{18}	2^{20}
$\ S_\mu\ _2/\sqrt{N}$	0.7798	0.7797	0.7797	0.7797
$\max_{\alpha \in (4N)^{-1}\mathbb{Z}/\mathbb{Z}} S_\mu(\alpha) /\sqrt{N}$	3.058	2.742	2.853	2.801
$(4N)^{-1} \sum_{j < 4N} S_\Lambda(j/(4N)) /\sqrt{N}$	1.946	2.084	2.219	2.346
(i) bound/ N	2.297	2.473	2.639	2.795
$N/((\text{grid max } S_\mu)(\text{grid mean } S_\Lambda))$	0.168	0.175	0.158	0.152

The first row reproduces $\sqrt{6/\pi^2} = 0.779697$ to four significant figures — the entries before rounding are 0.779764, 0.779725, 0.779686, 0.779699 — confirming the discrete Parseval normalization; the last row is only the corresponding grid diagnostic for route (ii), not an evaluation of the continuous product in Proposition 29. It sits near 0.16 and falls over the last three abscissae. Route (i) diverges from the trivial bound like $(\log N)^{1/2}$, as predicted. For reference the object itself is small: $C(N)/N = -0.0105, 0.0001, 0.0059, 0.0032$ at these N (`audit_circle_margin.py`).

6 Summary

- The design space of divisor-switch weights is described exactly by six finite rearrangements — Propositions 4, 5, 6, 9, 10 and 11 — with Proposition 13 a consequence of them rather than one of them: the per-modulus discrepancy is a dilate, the signs $\mu(k)$ cancel on the $\log k$ branch, and the untruncated object is a prime-supported weighted pair sum differing from $\tilde{r}(N)$ only by proper-prime-power terms.
- Theorem 20: on the explicit low-support class where the residual is controlled termwise by Bombieri–Vinogradov, the displayed triangle-inequality extraction incurs a prefactor larger than every fixed power of $\log N$, even when a homogeneous strong bound for T_w is granted. Thus that majorant does not certify a fixed log-power saving for $C(N)$. This is not a necessity theorem for the support condition, and evaluations through sign correlations are not excluded. The quantified prefactor is $\exp(c_1 \sqrt{(\frac{1}{2} + \delta) \log N})$ up to the implicit constant.
- Theorem 24: with any fixed power slack in the support, the separation survives $EH(N^{\theta_E})$ for every fixed $\theta_E \in (1 - \theta', 1)$. The power-scale factor $N^{1-\theta_E}$ can vanish only at the formal endpoint $\theta_E = 1$, which the theorem does not analyse.
- Proposition 26: polynomial weights, which lie outside the hypothesis of Theorem 20, are analysed through nonnegativity of Λ_D and the circularity of the most direct leading-term tuning. Their complete part is not proved to be $o(N)$ or to have a useful lower bound. This is a statement about what remains open, not an impossibility theorem; weights that are neither in the low-support class nor polynomial are not treated here.
- Proposition 29: two natural global Hölder pairings for $C(N)$ have zero margin, with Parseval supplying the floor. A major/minor-arc decomposition with bilinear treatment is a different mechanism and is not addressed.
- No new bound for $C(N)$, and hence no new Goldbach theorem, is obtained. What is closed is one explicit support-based triangle-inequality evaluation, not the full weight space.

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Disclosure statement

No potential competing interest, financial or non-financial, was reported by the author. No funding was received for this work. Generative-AI systems were used for derivation assistance, computation, formalisation, and drafting and revision of the manuscript. The appendix records the tools and the limits of the computational and formal checks. The author reviewed the final mathematical statements and takes responsibility for all claims.

Data availability statement

The code behind every figure printed here, the output it produced, and the Lean 4 development are openly available at <https://github.com/soke91/arithmetic-convolution-fields>, in the directory P2-no-go-divisor-switch/. Two percentages quoted in Measurement 8 are read off two printed columns of a cited result file rather than produced by a run, and that passage says which columns and at which N . No data derived from human or animal subjects were used; every quantity here is computed from the integers.

A Use of an AI assistant, and what checked what

Generative-AI systems were used extensively during the development of this work, including derivation assistance, code generation and execution, formalisation, and drafting and revision. These tools are not authors. The author reviewed the final mathematical statements, computational outputs cited in the text, and generated revisions, and takes responsibility for every claim made here.

Which tool, and how. The initial manuscript and computational development were prepared with Anthropic’s Claude, model Opus 5, run through the Claude Code command-line interface. It was used to derive and check identities, to write and run the scripts in the accompanying code, to write the Lean development, and to draft and revise the text. A later mathematical audit and revision used OpenAI ChatGPT (GPT–5.6 Sol) to check the scope of the no-go claim, the homogeneity of the weighted switch, prime-power bookkeeping, cross-references, and LaTeX compilation. Earlier stages used earlier Claude-family models; the repository does not record which. The author takes responsibility for the final statements and for checking the generated revisions.

The list below distinguishes the kinds of checks actually performed and their limits.

Machine-verified. The finite rearrangements that lay out the design space are formalised in Lean 4 against Mathlib, in the file `lean/GoldbachLean/Layers.lean` of the code packet: the exchange of the double sum along its inner variable that Proposition 10 performs, the nonnegativity of the layers it produces, and the sense in which the two ends of Proposition 6 are one expression. Each depends on `propext`, `Classical.choice` and `Quot.sound` and on nothing else; the axiom report is in the packet. The Lean statements take Λ , μ and $\log$ as arbitrary functions, so what is verified is that these identities use no arithmetic property of them — which is what makes them rearrangements rather than estimates.

Not machine-verified. Theorem 20 rests on Bombieri–Vinogradov, which is not part of the Lean development used here, so the analytic no-go theorem is not formalised. What is formalised is the finite algebraic scaffolding, not the analytic conclusion.

Checked by computation. Every figure quoted is produced by a named run in the accompanying code, with the two exceptions named in Measurement 8, which are read off two printed columns of a cited file. A check in the build refuses a printed figure that no cited run produces. Forty-three rules carry a verdict across the fourteen cited runs, and every one is reported with it, including the ten the runs did not meet.

Cross-read. The proofs and the computations were additionally read by separate AI sessions working without sight of each other’s findings, and several statements here are corrections those readings produced — among them the reference against which the cross- k gain is compared, which an earlier printing took to be a count of moduli. It is not a substitute for either of the checks named as absent above.

B Reproducibility

The code and the output it produced are at

<https://github.com/soke91/arithmetic-convolution-fields>

in the directory `P2-no-go-divisor-switch/`. Each entry below is run as `python code/<script>.py` from that directory and writes the file of the same name in `results/`. The figures here were produced under Python 3.12.10 and NumPy 2.2.5. That address is mutable, so a number checked against the branch head may have been recomputed since; the release named there pins the state these figures were read from.

Two files are in that directory without being named below: `lab_layer_tail.py`, whose output `lab_combined_modulus.py` reads, and `indep.py`, which `c02_flatsum_k1.py` imports. Run `lab_layer_tail` before `lab_combined_modulus`; the rest are independent of each other.

A *nonzero exit status is not a failure*. These runs carry preregistered diagnostic rules, and a run whose rule was refuted records that outcome in both its exit code and its stored output. The complete rule ledger, including negative outcomes, is retained in the repository release rather than reproduced in full here.

Script	Result file	Supports
<code>lab_dilate_identity.py</code>	<code>lab_dilate_identity.txt</code>	Proposition 4
<code>lab_positive_weights.py</code>	<code>lab_positive_weights.txt</code>	Proposition 5
<code>lab_weight_gap.py</code>	<code>lab_weight_gap.txt</code>	Measurement 8, $k \geq 2$ column
<code>c02_flatsum_k1.py</code>	<code>c02_flatsum_k1.txt</code>	Measurement 8, $k \geq 1$ column
<code>lab_direct_identity.py</code>	<code>lab_direct_identity.txt</code>	Proposition 9
<code>lab_layer_decomposition.py</code>	<code>lab_layer_decomposition.txt</code>	Proposition 10
<code>lab_combined_modulus.py</code>	<code>lab_combined_modulus.txt</code>	Proposition 11
<code>lab_direct_route.py</code>	<code>lab_direct_route.txt</code>	Measurement 15, residual and parenthetical list
<code>a1_bh_vs_b.py</code>	<code>a1_bh_vs_b.txt</code>	Measurement 15, the two lists
<code>audit_extraction_tradeoff.py</code>	<code>audit_extraction_tradeoff.txt</code>	Measurement 25
<code>audit_polyweight.py</code>	<code>audit_polyweight.txt</code>	Measurement 28
<code>audit_circle_margin.py</code>	<code>audit_circle_margin.txt</code>	Measurement 30
<code>audit_nogo_scope.py</code>	<code>audit_nogo_scope.txt</code>	Note 22
<code>audit_squarefull_admitted.py</code>	<code>audit_squarefull_admitted.txt</code>	the note after Proposition 6

No computation is used as a premise of any proof. Six of the seven statements of Section 2 are finite rearrangements and are verified as such to machine precision; Proposition 13 is the seventh and is not one of them — it carries an $O_A(N(\log N)^{-A})$ — so what is verified there is its identity part and not the error term. The measurements record sizes over the ranges stated in each statement.

The stored audit ledger includes negative as well as positive diagnostic outcomes. Eight of the fourteen cited runs refute at least one of their own preregistered checks; none of those checks is used as a premise of a proof. One finite-range diagnostic concerns the convergence $B_{\log}(K) + \mathfrak{S}(N) \rightarrow 0$ used in Proposition 13; its failure records only that the sampled range does not display the asymptotic rate. The exact rule statements, numerical outcomes, and exit codes are preserved in the cited result files for auditability.